Dual-Column Reading Order & Cross-Page Continuity Test

1. Left Column: Pipeline Bottlenecks

Traditional pipeline OCR tools split layout detection, formula recognition, and table extraction into separate models. When faced with two-column academic papers, heuristic sorting rules frequently scramble the reading order by merging lines horizontally across columns.

2. Right Column: R-SWA Causal Tracking

By feeding the 128-token causal sliding window continuously, Unlimited-OCR tracks its exact reading position from the bottom of the left column to the top of the right column, achieving a 7/9 win rate over DeepSeek-OCR 2 on reading order edit distance.